

Differential Gene Expression on SARS-CoV-2 Data

Lab Course Genome-Oriented Bioinformatics

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1 Introduction

Gene expression is the process by which the information encoded in a gene is synthesized in order to produce molecules such as proteins. Differential gene expression analysis seeks to identify genes whose expression levels differ significantly between experimental conditions. This helps in understanding the biological response to infections or other stimuli.

In this report, RNA sequencing (RNA-seq) data was analyzed to investigate host gene expression changes in response to infection by SARS-CoV and SARS-CoV-2. Using the DESeq2[1] framework, a performed DGE analysis was performed on datasets mapped with STAR[2] and HISAT [3] aligners to ensure methodological robustness. By identifying differentially expressed genes, this analysis provides insights into the molecular mechanisms underlying viral infections and host immune responses.

2 Materials and Methods

2.1 Datasets

Two separate gene count files from an RNAseq experiment of COVID-19 data were used for analysis. These files were generated using two different RNA-seq read alignment tools, STAR[2] and HISAT2[4]. The gene count files produced by these aligners will be referred to as *star* and *hisat*, respectively, throughout the rest of this document.

The samples in the dataset are categorized into three main experimental conditions:

- **S2** - SARS-CoV-2 infected cells
- **S1** - SARS-CoV infected cells
- **mock** - Mock-infected cells

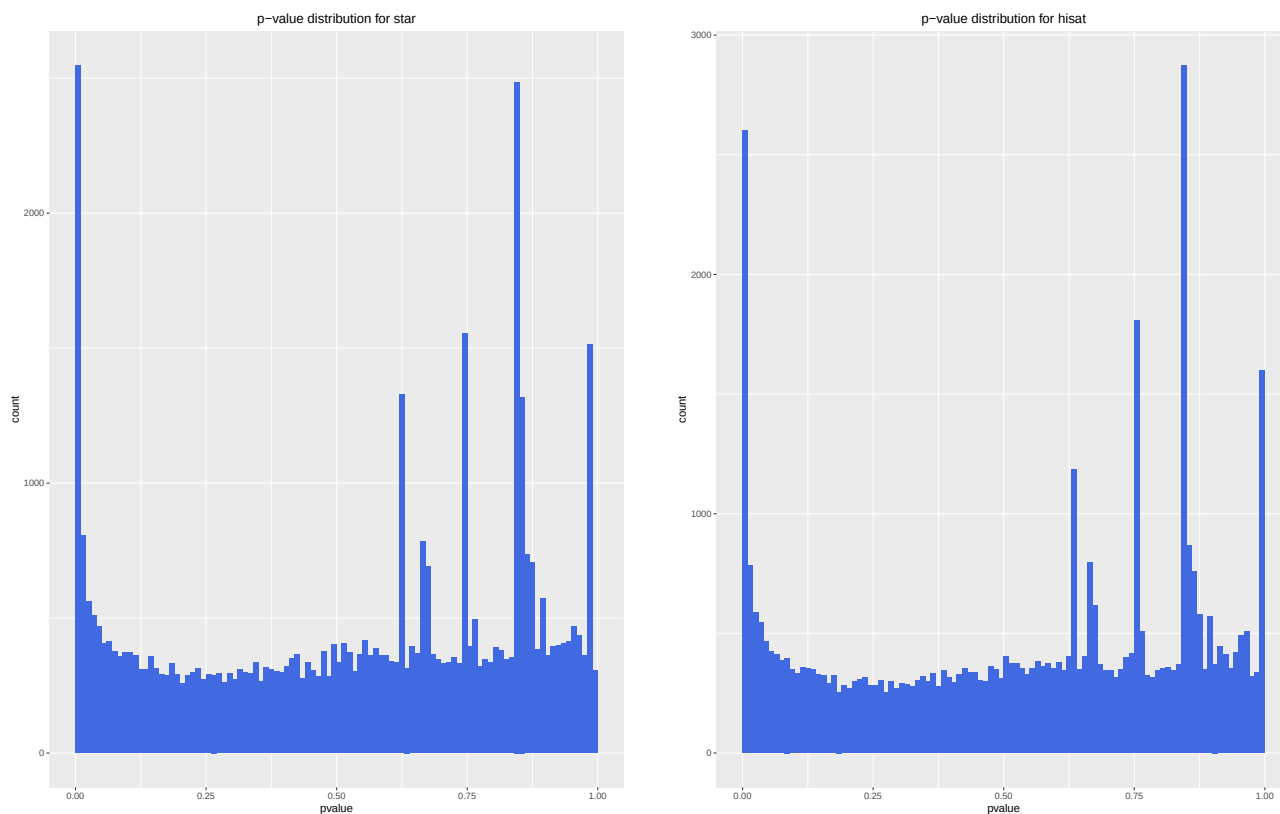
For both **S1** and **S2** infections, several time points post-infection were measured: 4 hours (4h), 12 hours (12h), and 24 hours (24h). For the **mock** condition, two time points were measured: 4 hours (4h) and 24 hours (24h).

2.2 Differential Gene Expression Analysis

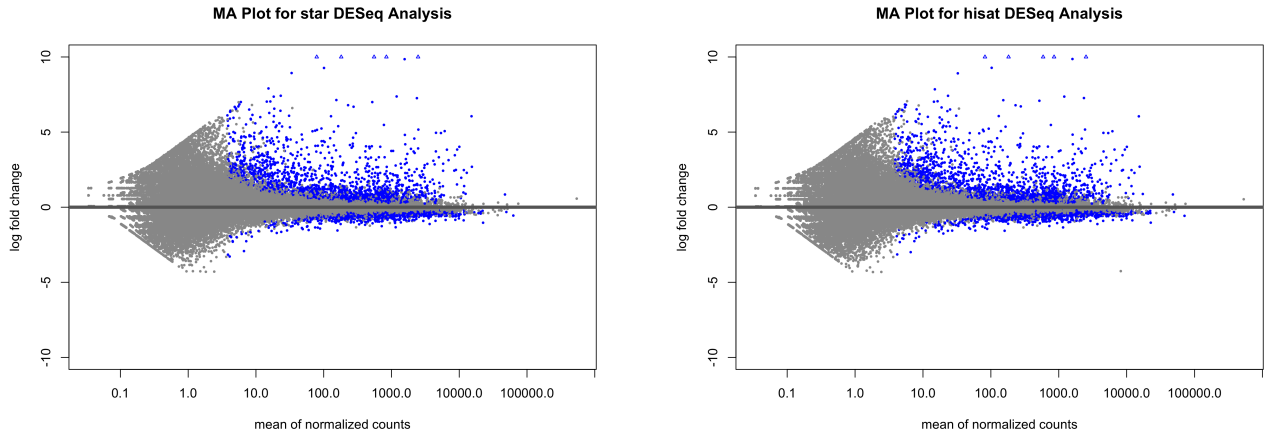
Differential gene expression analysis was performed using the R package DESeq2[1]. The analysis was performed separately for the *star* and *hisat* datasets. The analysis was performed for each time point separately, comparing the gene expression levels between the infected and mock-infected cells. The analysis was performed according to the steps described in Modern Statistics for Modern Biology, Chapter 8[5].

3 Results

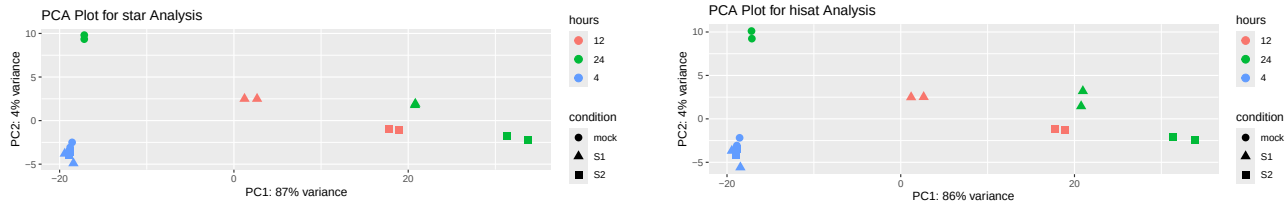
The histogram displays the distribution of p-values for differentially expressed genes. A clear peak near zero indicates a significant number of differentially expressed genes, suggesting strong signals of gene expression changes. A smaller peak at high p-values represents genes likely not differentially expressed. Approximately 2,600 genes are in the lowest p-value bin (STAR and HISAT), with minimal differences in overall distributions.



The MA plot visualizes the relationship between the mean expression levels (x-axis, log2-transformed) and the log2 fold change (y-axis). Points represent genes, with significant genes (adjusted p-value < 0.1) highlighted in red. The majority of data points cluster around zero, indicating no significant change, while a subset shows large fold changes. No meaningful differences were observed between STAR and HISAT in terms of significant genes or patterns.

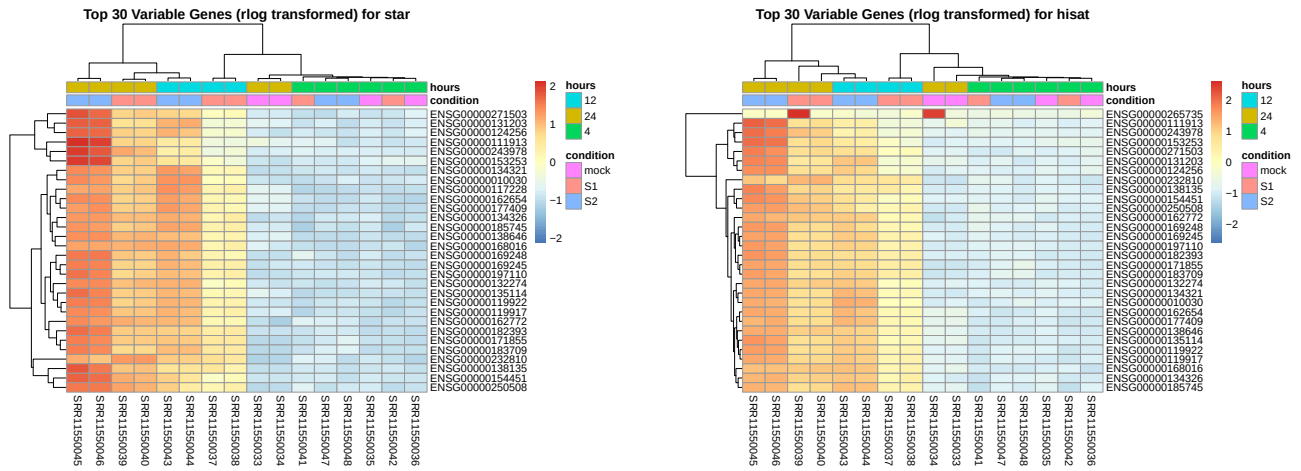


The PCA plot shows the variance in the dataset explained by the first two principal components (PC1 and PC2). PC1 accounts for approximately 86-87% of the variance, while PC2 accounts for 4%. Initially, all samples cluster together near the origin (early time points, indicated in blue). Over time, samples spread along PC1, reflecting time-dependent expression changes. The mock samples remain relatively unchanged, while S1 and S2 diverge along PC1.

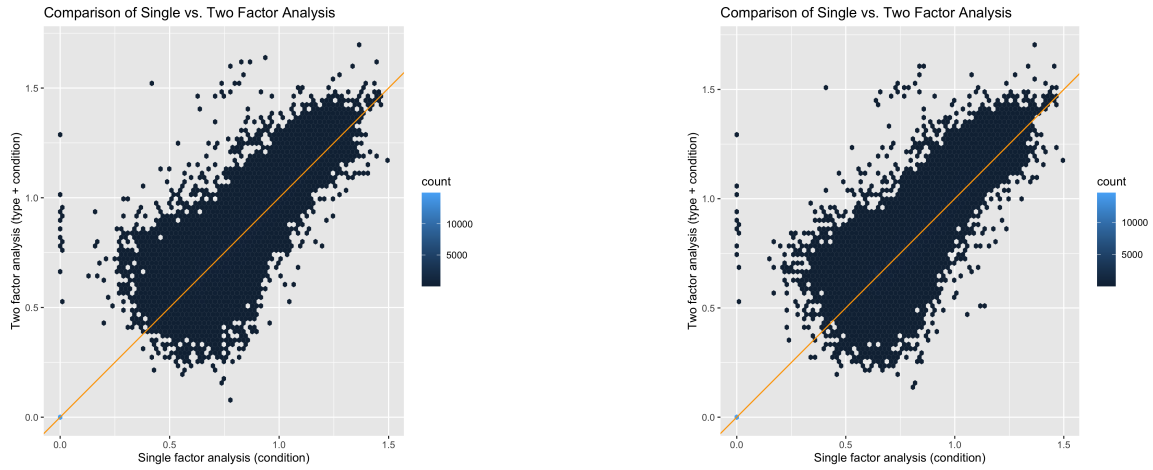


The heatmap visualizes the top 30 variable genes, identified based on the highest variance in expression across samples. Variance-stabilized counts were calculated using the rlog transformation, which stabilizes variance across genes and accounts for differences in sequencing depth. The mean expression value for each gene was subtracted from its rlog-transformed counts to center the data. This normalization highlights deviations from the mean, with positive values indicating higher-than-average expression and negative values indicating lower-than-average expression. Hierarchical clustering was applied to group genes and samples based on expression similarities. The clustering revealed:

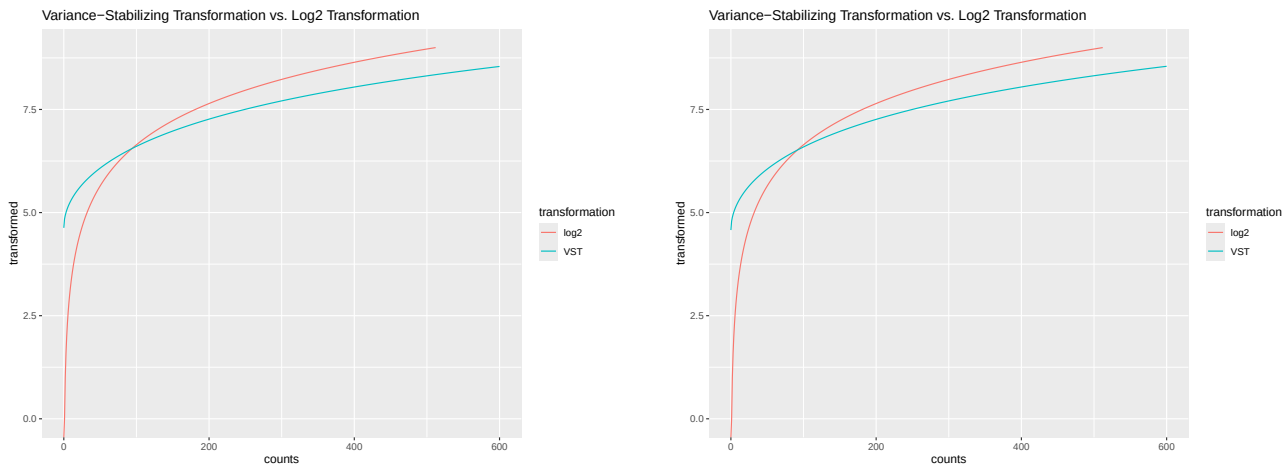
- At 4 hours post-infection, mock, S1, and S2 samples cluster together, reflecting minimal transcriptional divergence.
- At 12 and 24 hours, samples cluster by time point, with S1 and S2 grouping within each time cluster, indicating time-dependent transcriptional changes.
- Notable anomalies include a uniquely variable gene in mock and S1 at 24 hours (STAR dataset) and heightened variability in S2 at 24 hours, suggesting a distinct response.



This plot compares p-values from single-factor models (condition) against those from two-factor models (condition and type). The axes represent a transformed p-value scale to compress the range for better visualization. The distribution exhibits an "eggplant-like" shape, indicating some differences in the models' sensitivity. A subset of genes is more significant in one model than the other, reflecting model-specific dynamics.



This graph compares the VST and log2 transformations for one sample. Both transformations converge for large counts, while VST stabilizes variance better for small counts, avoiding the steep slopes seen with log2 transformation. The curves intersect around a count of 100, highlighting differences in transformation behavior. The results are consistent across datasets.



4 Discussion

Differential gene expression analysis allows us to uncover the expression changes that are induced by pathogens. In this case, RNA-seq data helps us understand the cellular responses to SARS-CoV and SARS-CoV-2 infections across time points. The results show a clear time difference, with gene expression patterns diverging significantly from mock-infected controls as time progresses. The different mappers used, STAR and HISAT, showed consistent results. The observed clustering patterns, particularly the temporal separation of S1 and S2 conditions, reflect shared and distinct host responses to these closely related coronaviruses. Notably, the heightened variability in S2 at 24 hours suggests a unique transcriptional response to SARS-CoV-2 infection that requires further investigation to explore how it differs from SARS-CoV infection.

References

- [1] Michael I Love, Wolfgang Huber, and Simon Anders. “Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2”. In: *Genome Biology* 15.12 (Dec. 2014), p. 550.
- [2] Alexander Dobin et al. “STAR: ultrafast universal RNA-seq aligner”. In: *Bioinformatics* 29.1 (Oct. 2012), pp. 15–21. DOI: <https://doi.org/10.1093/bioinformatics/bts635>.
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